Manual

How to Operate the MES Operation Center

Version 1.34

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Contents

1 Operation of the MES Operation Center

1.1 General notes on the document

This document describes how to operate the MES Operation Center (MOC). This manual focuses on a detailed presentation of the general functions, which are available for all applications. The different applications are described in the respective application manuals, which can be opened by clicking the F1 button or using the link in the application package manual.

1.2 Requirements for using the MOC

For the MOC installation, the PC must meet the following requirements:

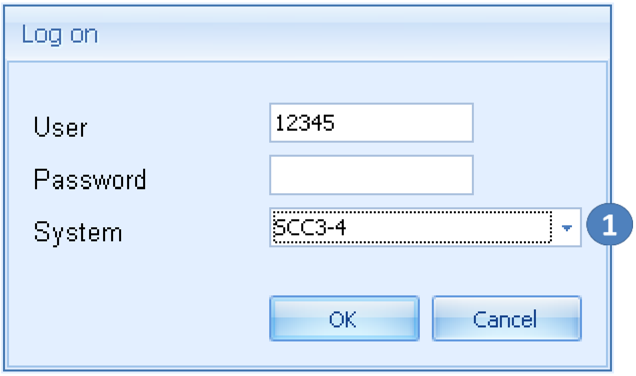
- Installed operating system: Windows 7 or higher
- .NET Framework 4.5.2
- PDF Viewer to use the MOC help

When being started, the MOC checks whether these components are available or not.

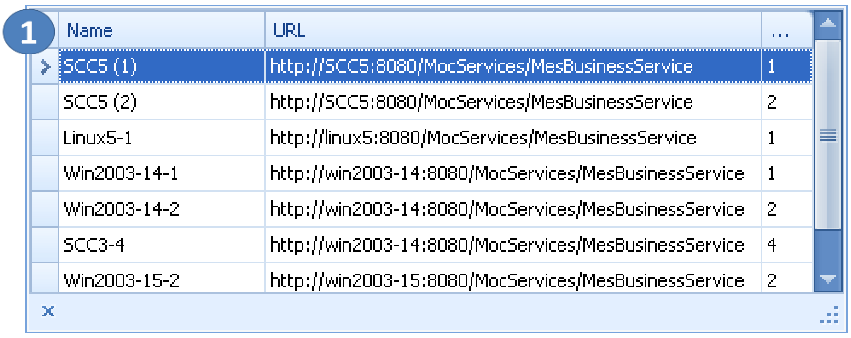
2 Getting started

2.1 Start and exit the MOC

Start the MOC via the Windows start menu or use a link on the desktop. While being started, the MOC start screen is displayed, visualizing the loading process. Once all necessary components have been loaded, the MOC start screen closes and the MOC desktop with its login window appears. Enter the user name and password you have received from system administration.



If the MOC installation is a multi-system installation, you can select the system you want to log on to from the *System* combo box.



Name	URL	...
SCC5 (1)	http://SCC5:8080/MocServices/MesBusinessService	1
SCC5 (2)	http://SCC5:8080/MocServices/MesBusinessService	2
Linux5-1	http://linux5:8080/MocServices/MesBusinessService	1
Win2003-14-1	http://win2003-14:8080/MocServices/MesBusinessService	1
Win2003-14-2	http://win2003-14:8080/MocServices/MesBusinessService	2
SCC3-4	http://win2003-14:8080/MocServices/MesBusinessService	4
Win2003-15-2	http://win2003-15:8080/MocServices/MesBusinessService	2

The OK button logs you in. Note: User name and password are case sensitive.

The Cancel button terminates the MOC and cancels the login process.

After successful login process, the system reads the user's authorizations and releases the system menu and the start menu. We recommend changing the password given by the administrator when logging in to the MOC for the first time. To do so, select *File* → *Change password* in the system menu.

You can start and run the MOC several times on a PC. Each started MOC instance can access another system. Consequently, you can access the integration and production/live system or monitor the systems of two different factories at the same time.

Log off

Select this option in the system menu to log off from the current system: *File* → *Log off.* Then you can either log on another user or you can change the target system.

Exit

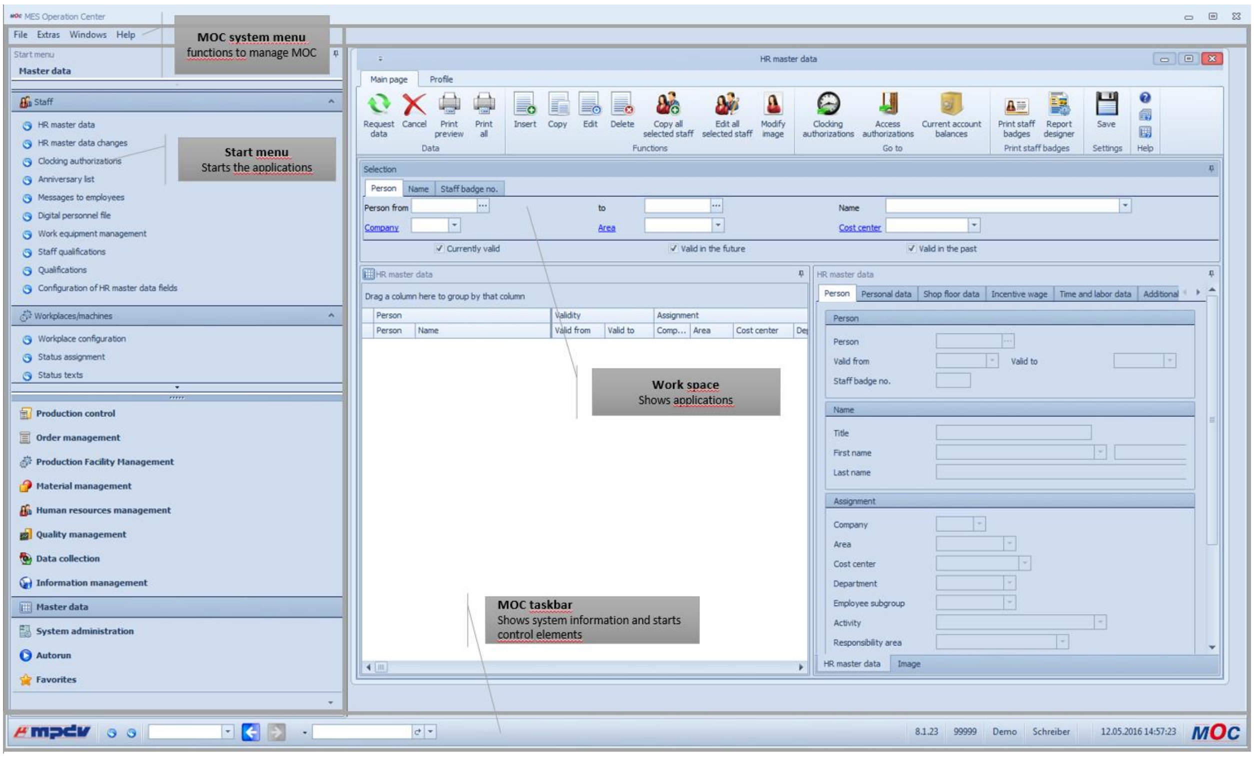
Select this option to exit the selected MOC instance: MOC system menu → *File* → *Exit.*

Language check

With each MOC login, the system checks whether the language last selected is licensed. If the language is not licensed, an error message is displayed informing about the missing license. The MOC contents are then displayed in the fallback language (English). Change the language in the system menu. Go to: *Extras* → *Configuration*.

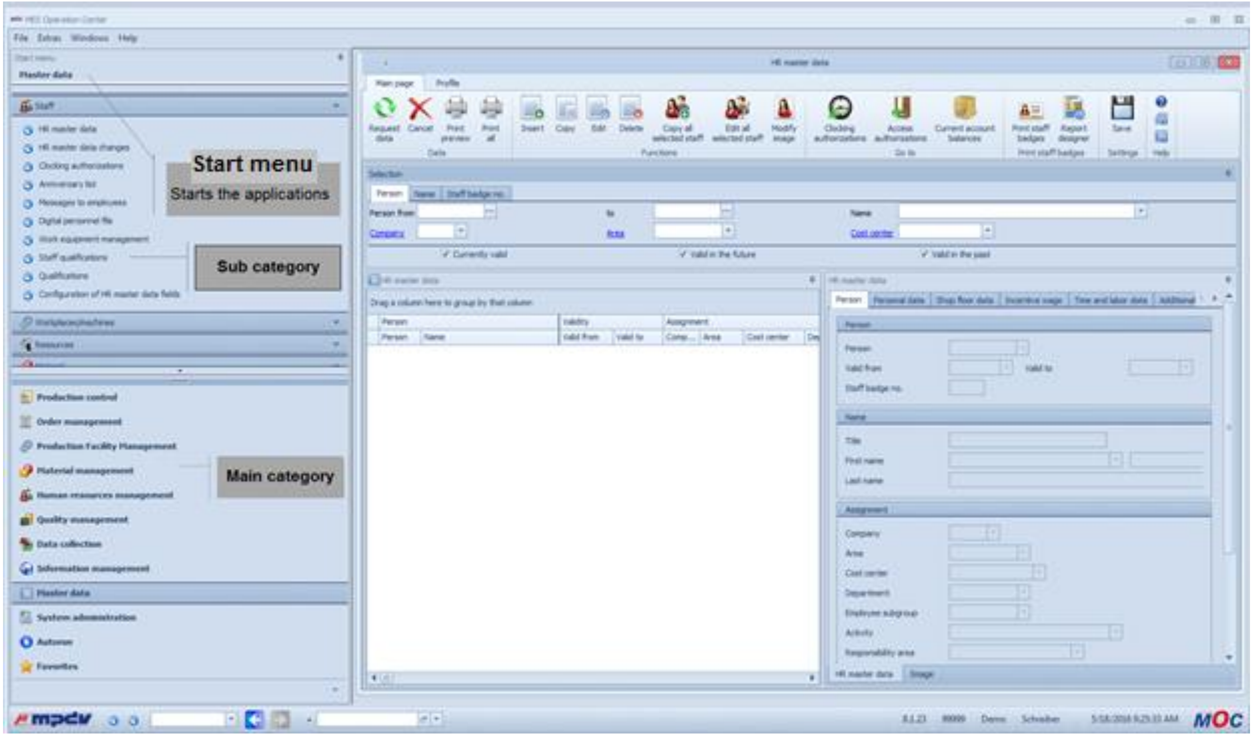
2.2 The MOC desktop

The MOC desktop is the frame application including all control functions, such as the start menu and the quick launch bar. The MOC desktop also shows the different applications.

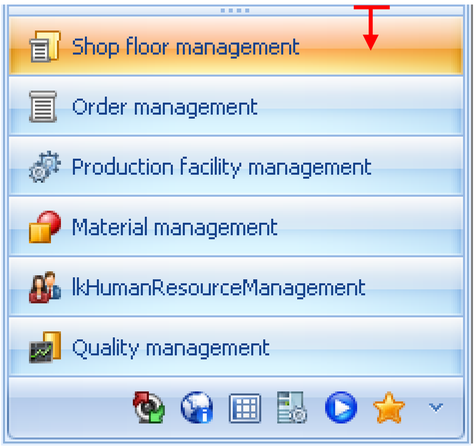


2.3 MOC Start menu

The MOC start menu provides access to the MOC applications. By default, the function calls provided in the MOC start menu are based on the VDI 5600 role definition. The user can change the menu according to the individual requirements. See section "4.1.5 Change the start menu" and "4.2 Menu editor".

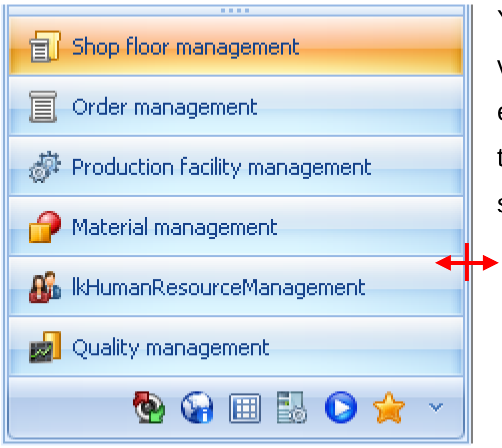


Minimize main categories to the footer of the start menu



The start menu is split horizontally into main categories and sub categories including their menu calls. You can use this splitter to minimize the main categories to the footer of the start menu. To do so, use the splitter to drag the area of the main categories down. Then the main categories are displayed as icons. If the mouse touches one of the icons the name of the category is displayed in a tooltip.

Change the width of the start menu



You can change the width of the start menu by moving the vertical splitter. If the selected width is too small, the menu entries cannot be displayed completely any more. In this case, a tooltip appears if the mouse points at a menu entry for some seconds.

Dock the start menu

By default, the start menu is docked to the MOC desktop on the left-hand side. But you can also dock the start menu to the top, the bottom or the right-hand side. To do so, click the title bar of the start menu and drag it out of the docking position. The possible docking positions are now displayed for better overview. You can also leave the start menu undocked on the MOC desktop or outside of the frame application.

Hide the start menu / pin

You can hide the start menu in order to clear space for applications on the MOC desktop: Click when the start menu is undocked. If the start menu is docked, you can also use the pin option . Once clicked , the icon is displayed in a horizontal position and the start menu is hidden automatically as soon as the mouse has left the menu.

Hide categories in the start menu

This button in the footer of the start menu opens a menu to configure menu categories. Here, you can hide or show main categories.

Filter start menu by function profile

You can select any function profile in the MOC taskbar. The system then filters the menu according to the selected function profile. Only the menu items are displayed, which are included in the function profile.

Manage favorites in the start menu

You can define applications as favorites to adjust the function calls according to your personal requirements. Select an application in the start menu, open the context menu and choose *Add to favorites.*

The selected favorites are then contained in the *Favorites* category. 

You can also remove entries from the *Favorites.* Select *Remove* in the context menu.

Autorun

Applications stored in the *Autorun* menu are opened automatically when you start the MOC. Select an application in the start menu, open the context menu and choose *Add to autostart.*

MOC authorization

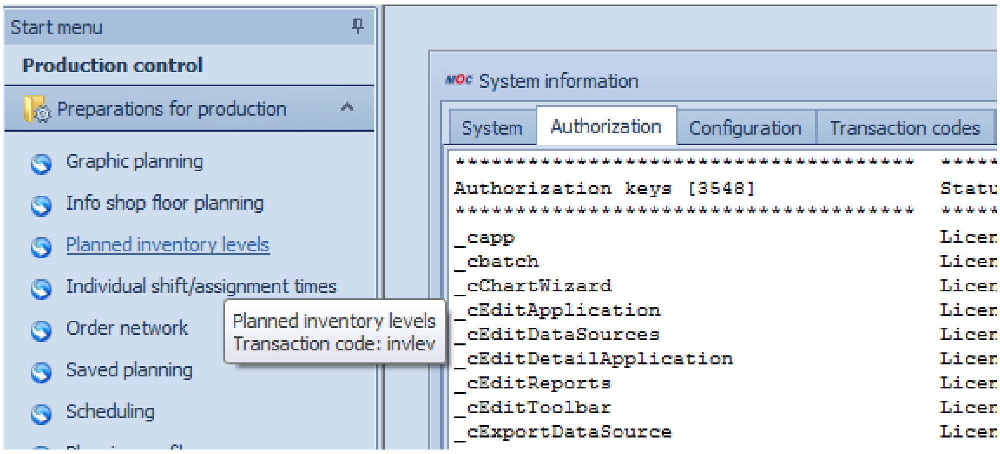
HYDRA 8 MOC authorization works as described below. The authorization process affects the following MOC objects:

- Applications (e.g. *workplace overview).*
- Detail applications (e.g. *cycle view* in the application *workplace overview).*
- Functions you can start (e.g. the *Edit* function in the application *Units).*
- Fields and field groups in detail applications.

You first assign authorization keys to the listed objects. To release the objects, you require a function authorization or a license.

Authorization of applications

If an application is grayed out in the start menu, this application is neither authorized nor activated. Install licenses or create function authorizations to **activate** the component. The tooltip of the grayed out application tells you whether a license or function authorization is missing for the application.



2.4 The MOC system menu

The MOC system menu includes functions to manage the MOC. This is a fixed menu that can neither be changed by the user nor by the system administrator. The functions provided in the system menu are explained in section 4 System functions.

2.5 The MOC taskbar

The MOC taskbar is located at the bottom of the MOC desktop. The taskbar contains control functions such as the forward/backward navigation or transaction calls and shows also system information, such as the currently used system, the current user, etc.

Call applications using transaction codes

You can call applications using the input field of the taskbar that is to the right. When you click the input field, a list opens showing the transaction codes and names of all available applications. Enter a search term to filter the list. A matching list entry is highlighted. To select the relevant entry, press the *Enter* key or click the mouse. The quick launch shows the application that was last opened. Click the *Repeat* button to restart this application.

Navigate forward and backward

When working with the MOC, several applications are usually opened at the same time. Use the navigation arrows to follow the course of application calls and to activate the relevant applications. Consequently, you can go to applications, which have already been opened, or reopen applications, which have been closed. You can also display or change the currently opened applications using the *Windows* option or change the active application using Ctrl + Tab.

Show open applications

The drop-down option next to the navigation arrows in the MOC taskbar opens the history. This list also shows all applications that are currently opened. Use the shortcut Ctrl+TAB to go through all open applications.

System information

The MOC taskbar shows the following system information on the right-hand side:

- Version: Version of the MOC software in use.
- Customer number: Customer number that is set in the used instance.
- User: User that is currently logged in to this MOC instance
- System: System the used MOC instance is connected to
- System time: Date and time.

2.6 MOC help functions

The MOC help consists of three components. Click the F1 key of your keyboard to open the help function. Then the MOC opens the help function that corresponds to your selected item. If you select a detail application, for example, and press the F1 key, the help function matching this detail application opens.

You can also open the help function in every application by clicking the respective toolbar button:

Help on operation

Click this button to open the help file describing how to operate the MOC. The document is entitled “MOC_CC.pdf” (this document). The document describes how to use the MOC in general and applies for all applications.

Help on application

The function *Help on application* opens the manual that describes the application from which the help file was requested. The application manual explains how the application works. The documentation also includes all detail applications.

Help on detail application

The function *Help on detail application* opens the application manual at the section/bookmark where the respective detail application is described.

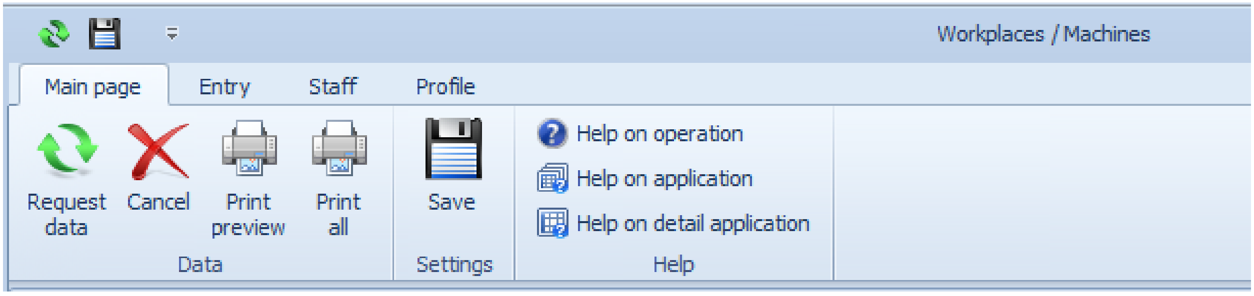
3 Structure and use of applications

3.1 Overviews and evaluations/reports.

Each MOC application has been designed and implemented for special activities. However, all applications have the same structure.

The toolbar

The toolbar contains all function calls of an application.



The toolbar has one or more tabs. The above example shows the tabs *Main page, Entry and Staff.*

Every tab is divided into categories. The above example shows the *Entry* tab with its categories *Data, Settings* and *Help.*

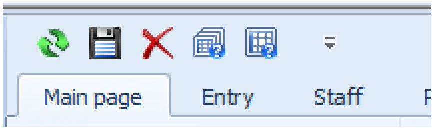
The toolbar is context sensitive. This means that for each detail application selected, the tabs assigned to this detail application are displayed. But the application functions of the categories *Data* and *Help* are always available regardless of which detail application is selected.

You can hide the toolbar by double-clicking a tab or opening the context menu.

In the same way, you can show the toolbar.

Quick launch bar

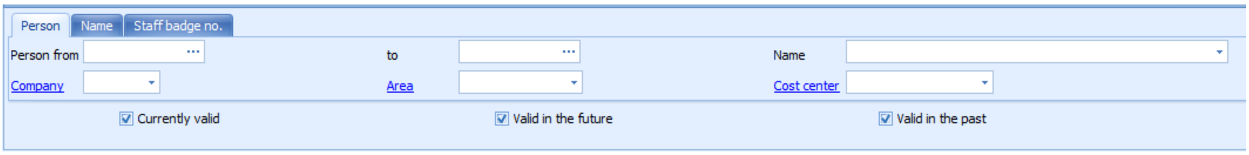
For each application, you can define individual application favorites that are displayed in the quick launch bar of the application. The user can use these functions even if the toolbar is hidden.



Choose the *Add to quick launch bar* option in the context menu of the function call to qualify a function as an application favorite. Consequently, the icon to start the function is inserted at the end of the quick launch toolbar. Choose the *Remove from quick launch bar* option in the context menu of the icon to remove an icon from the quick launch toolbar.

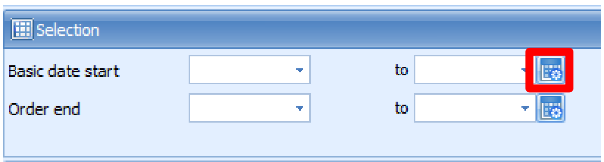
Selection pane

The selection pane includes filter criteria that are specific to each application. Each selection pane can contain one or more tabs with filter criteria.

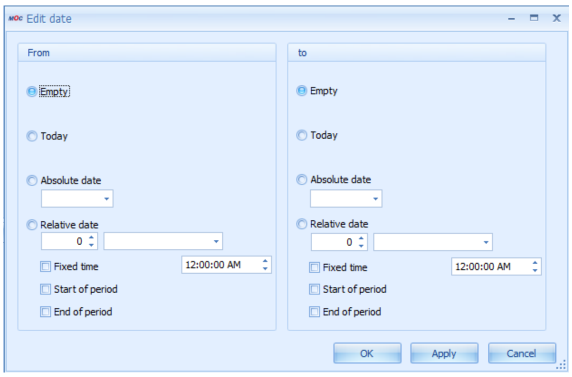


By default, each tab shows a maximum of two rows. You can access the other filters via the scrollbar to the right of the selection pane.

With some selection criteria, further, more detailed selection criteria can be called.

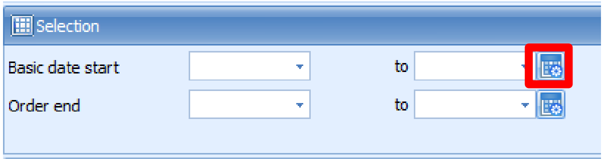


In the dialog that opens, further selection criteria are available. Example:



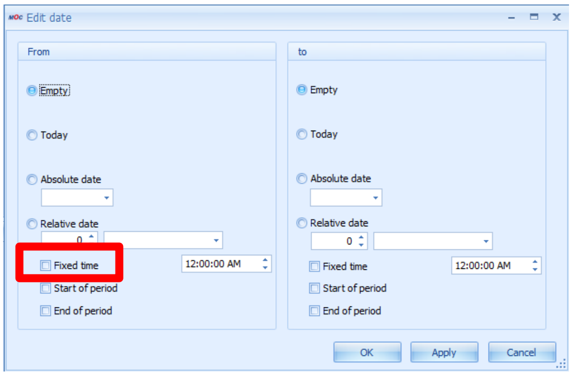
The above dialog is a standard software component. The dialog provides a predefined selection of selection criteria. The same standard software components are provided in different applications. The possible selection criteria are only processed as part of the selection if the relevant value is provided in the selection field of the application's selection pane.

Example:



The field *Basic date start* only provides a date selection.

The standard software component also permits a selection by "Fixed time".



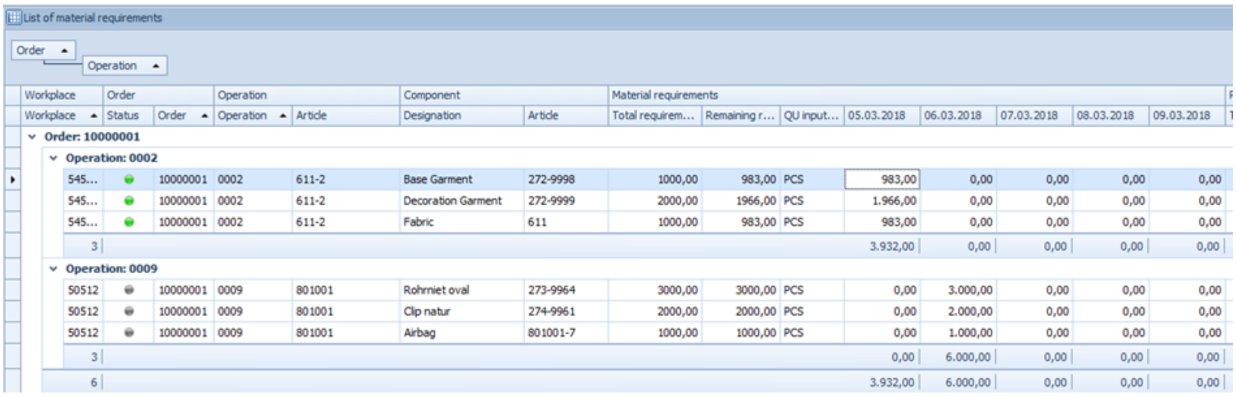
The selection field *Basic date start* does not provide the field *Fixed time*. A selection by *Fixed time* is therefore not possible.

Group columns for selection

You can group columns on the MOC and create a tree structure. The tree structure can sometimes include a drill-down functionality.

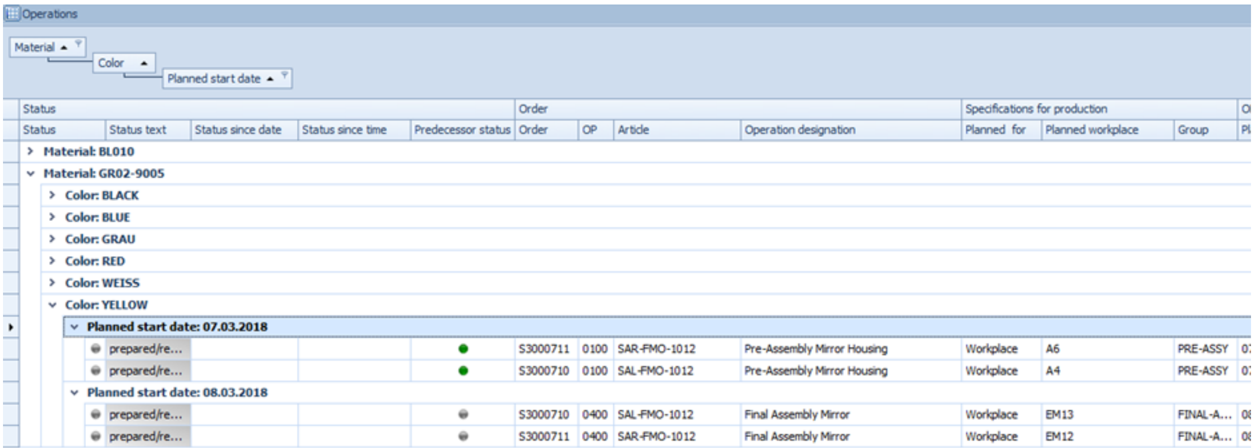
This kind of layout for selection criteria is helpful in many applications because this layout is flexible and provides many different combinations.

For example, material can be displayed in a selection by order and then by operation including also totals values (see screenshot).

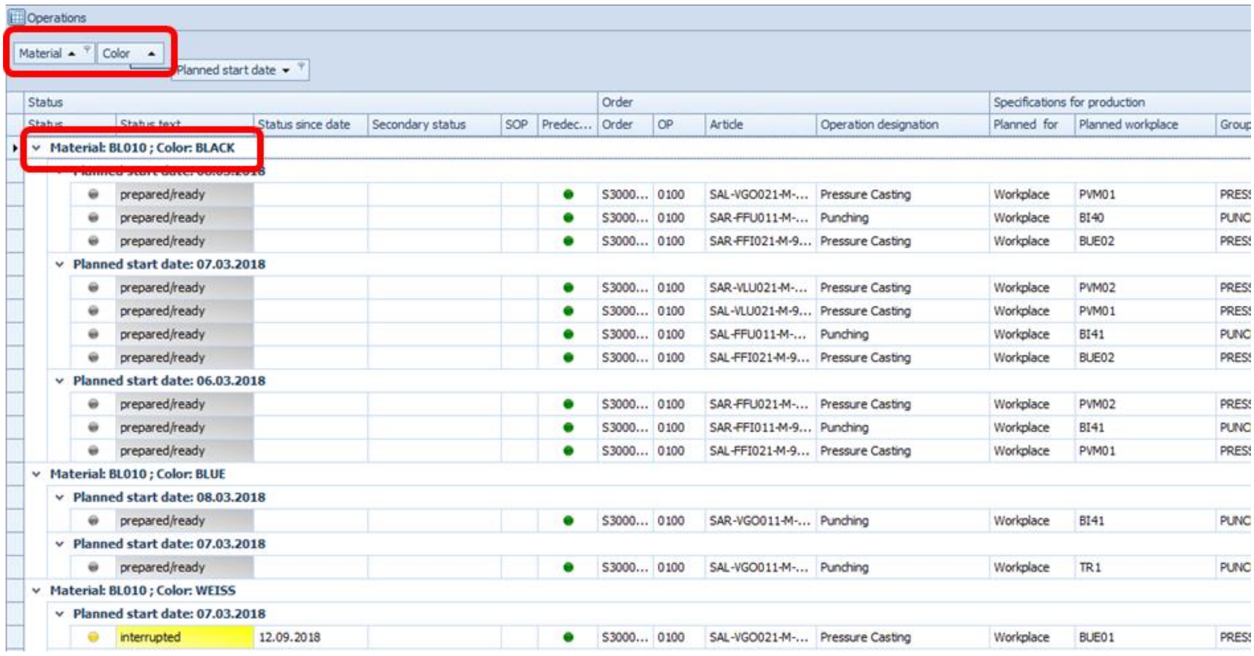


In other cases, it is helpful to generate one selection key from two (or more) columns in order to get a flat structure of selection criteria. When you next use the MOC application, you need less clicks to open the required data records and the possible combinations of column entries are immediately visualized as one selection key.

Standard structure:



Flat structure via column combination:



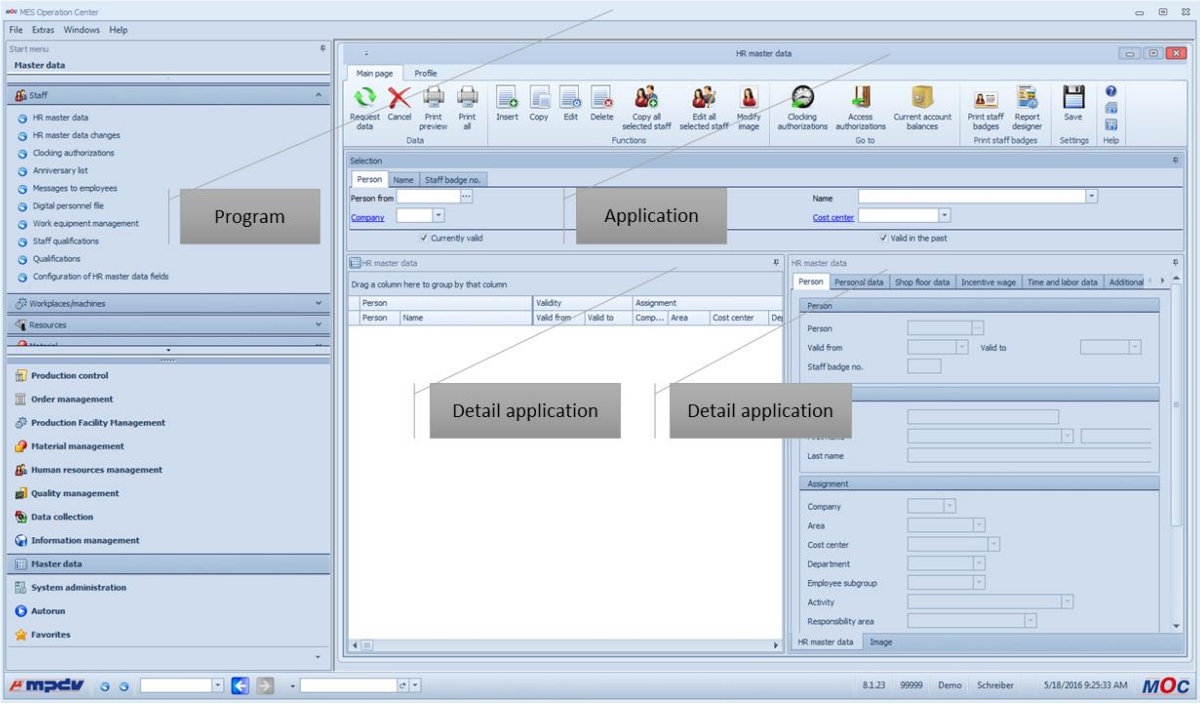
The context menu does not provide any command for the flat structure of the selection criteria. But you can use drag and drop and press the CTRL button at the same time to group the selection criteria.

Request data

When you have selected the filter criteria for an application, you can request data: make the relevant function call or press the ENTER key of your keyboard.

Detail applications

The functions of an application are included in one or more detail applications. If an application consists of more than one detail application, they are normally linked with each other. Examples of detail applications: tables or charts.



Save application layout

If you change the default application layout, for example, by adding or removing columns, changing the alignment of categories and columns, or by making other configurations for detail applications, and you want to keep these changes, you have to save this layout of the application. To do so, click *Save* in the *Settings* category of the toolbar.

Open an application several times

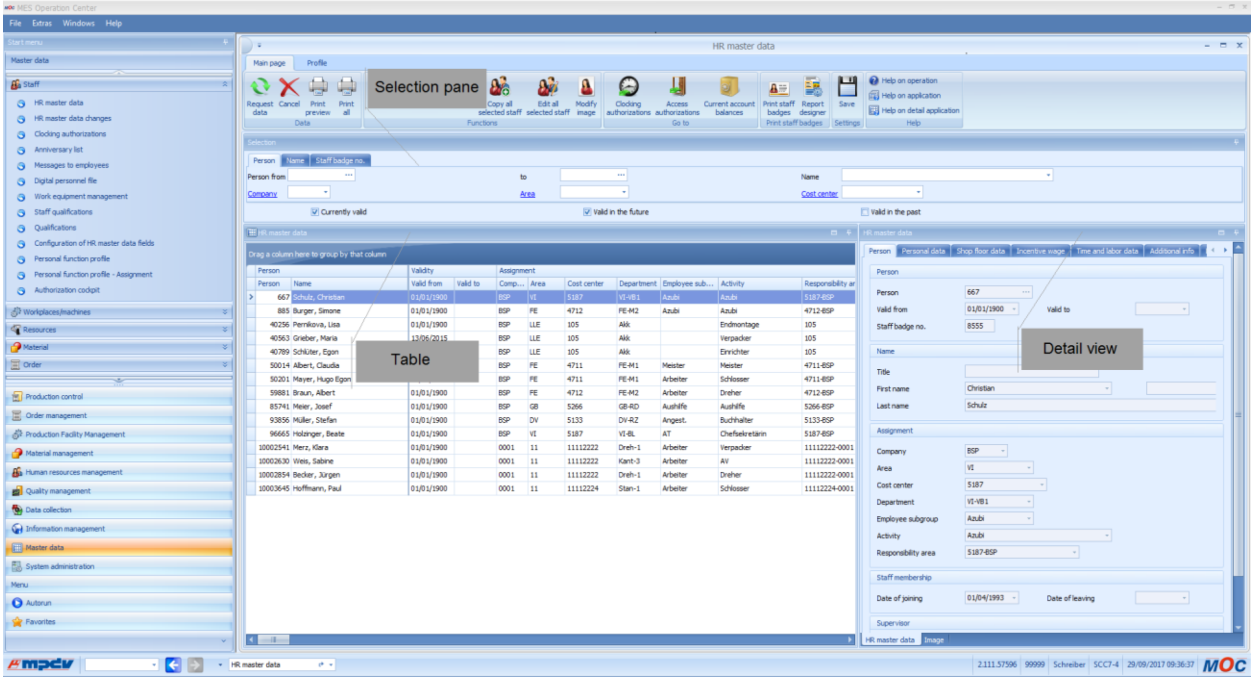
You can open most of the applications of an MOC instance multiple times. Just open the application once more using the menu or the transaction code.

3.2 Editing functions

The sections below describe how the editing functions work.

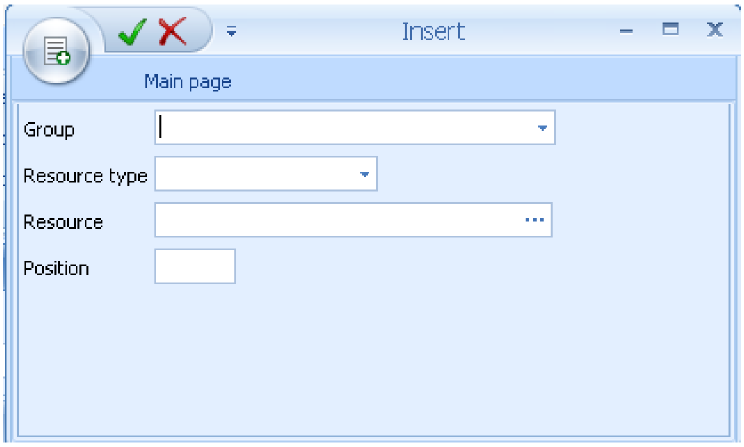
3.2.1 Main view of editing applications

Use the *Functions* category of the toolbar to call editing applications.



The main view of editing applications provides the toolbar to start functions and the selection pane – just as all other applications. In addition, the editing application shows a table of already existing entries and a detail view of the selected data record.

3.2.2 Insert new data records



Select *Insert* to add new data records in an editing application. The available input fields depend on the selected editing function. Click the green check to close the editing dialog. If you want to exit the dialog without saving data click the red cross.

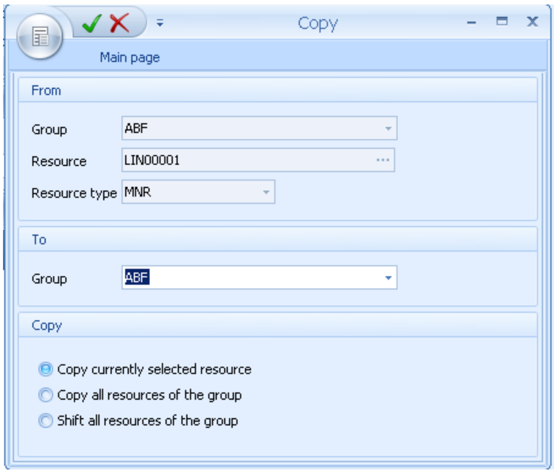
3.2.3 Copy a data record

There are two methods to copy data records. The method depends on the relevant application:

Data record as template

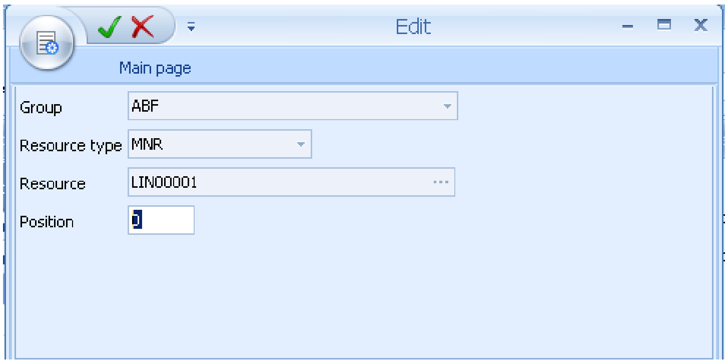
When you copy a table entry, the data record selected in the table is used as template for the data record to be inserted. You can edit the properties and close the dialog by clicking "OK".

Copy several data records



Use this option to copy several data records from one object to the next. The object of this example is the "group". The available copy options depend on the application that is selected.

3.2.4 Edit a data record



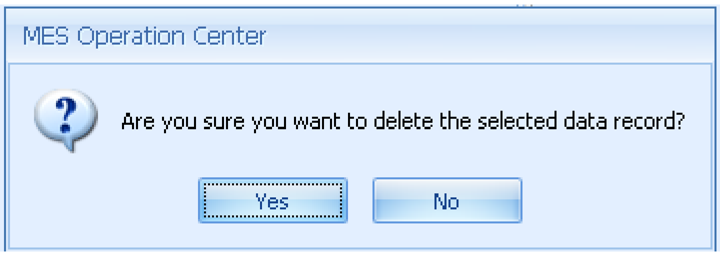
When you edit a data record, you can change some properties of the data record - in this example you can only change the "position". You cannot change the key values of the data record - in this case "group", "resource type" and "resource".

Click "OK" to confirm the dialog.

3.2.5 Delete data records

When you delete data records, you can either delete the selected data record or several data records. If you want to delete several data records at once, select the relevant records and click *Delete.*

If you select several data records, the system sends a confirmation prompt you have to confirm in order to delete these data records. The confirmation prompt should prevent you from deleting multiple data records unintentionally.



3.3 Operation of applications

3.3.1 Selection profiles

If you use evaluations/reports several times, it is helpful to save the frequently used filter criteria for evaluations/reports. For this reason, each application supports an individual management of such profiles. This means: You can create profiles for each application and reuse these profiles every time you open the application.

Create a selection profile

Go to the *Profile* tab of the toolbar to create a selection profile. Enter the name of the profile to be created in the combo box. Click the *Save* button to save the criteria that are currently set in the selection pane under the specified name.

Select a selection profile

You can select a selection profile from the combo box at any time. Once you have selected a selection profile, the selection fields are completed with the values stored in this selection profile.

Change a selection profile

You can change an existing selection profile by saving the changed selection criteria under the already assigned name.

You can overwrite global selection profiles with the function authorization "syspadm".

3.3.2 MOC table functions

You can change the tables on the MOC according to your requirements. You can use the functions described in the sections below with all tables (with very few exceptions).

The settings made for a table are only valid for the detail application where the settings were made. You cannot apply these changes to all tables in the system. To save the settings made for the tables and to re-use the table filter criteria, you must save the application settings (click *Save* in the toolbar).

Sort table data

Click the table header to sort table data in descending order. If you click the table header once more, data is sorted in ascending order. The selected sorting option is shown.

You can sort data by several columns: Press the *Shift* key of your keyboard after sorting the first column. Then click the other column headings by which you want to sort. You can start sorting as follows:

1. Right click the column name.
2. Select the option *Sort descending* or *Sort ascending.*

Group data in the table

You can group table data if the *group by* area is shown. If the *group by* area is not shown, you can show the area via the context menu of the table header (*Show/Hide group by box*).

To group by a column, click the column header and drag it to the *group by* area. Multiple grouping is also supported. If you have grouped table data, you can expand or collapse all groups. Open the context menu of the grouped elements and select *Full collapse* or *Full expand.*

Copy table data

Use the shortcut *CTRL + C* to copy the values of the currently selected table row. Use the shortcut *CTRL + ALT + C* to copy the currently selected table cell.

Optimum column width (best fit)

Select the option *Best fit* in the context menu of the table header to adjust the column width of the selected column to the optimum width. In this case, ”optimum” means that the column is as wide as the largest entry in the selected column.

Optimum column width (all columns) / Best fit (all columns)

Click this function to adjust all columns to the optimum width.

Change column width

You can also change the column width using the mouse, i.e. move the space between two cells to the left or right.

Show and/or hide columns and entire categories

Use the context menu function *Select columns* to show and/or hide individual columns and entire categories. For this purpose, select the function in the context menu and then drag the required columns and/or categories from the table to the pool or from the pool to the table.

Change the sequence of columns and categories

Also use the mouse to change the display order of columns and categories. To do so, drag the column or category you want to move and drop it at the required location. The system will indicate the location to which the column and/or category will be allocated when you release the left mouse button.

Filter table data

If you have already requested data, you can filter the data displayed in the table by entering individual criteria. To do so, you can use the AutoFilter function in the table header. Click the filter icon  in the column where you intend to set the filter and select the required filter option from the list; i.e. you select one of the values available in the table or compose a combination of values in the *user-defined* filter.

You can also set several filters in different columns. The table footer indicates that the table has been filtered and also shows the filter criteria. Select the function *Edit filter* on the right of the footer to open the filter editor. Use the filter editor to create complex filter criteria across all columns. You may also open the filter editor via the context menu of the table header.

Search box

Open the context menu of the table header and select the option *Show search box.* This option provides a search box in the table. Use this box to quickly search and/or filter the requested data. Simply start typing in this box and the system will only show those rows matching the data you typed. The more characters you enter, the more you narrow down the result.

Filter row

Open the context menu of the table header and select the option *Show filter row.* This option provides an additional row that is displayed below the table header. You can enter a search term in any column, and the system will narrow down the displayed rows appropriately. The system supports wildcards. You can also combine search terms in various columns to restrict the search result.

Show column totals for group

Open the context menu of the table header to select the *Show column totals for group* option. This option enables you to show or hide a totals row for each group *(subtotal).*

This option is only available if a totals row is displayed in the application. The totals row per group is only visible once you have saved and reopened the application. You cannot show/hide the totals row according to your specific requirements.

Fix category (freeze)

Open the context menu of the table header and select the *Fix category* option (freeze) to fix/freeze the required categories on the left-hand side of the table. The fixed/frozen categories are not affected by scrolling; they remain static.

3.3.3 How to use simple charts

Some reports support charts that depend on the selections made. This means: the charts of the different detail applications are based on the data records selected in the relevant table and only refer to these selected data records.

If you change the selected area, the chart presentation changes as well. Hold down the shift key to select areas in the table. Use the CTRL key to select specific data records.

3.3.4 Pivot tables in combination with pivot charts

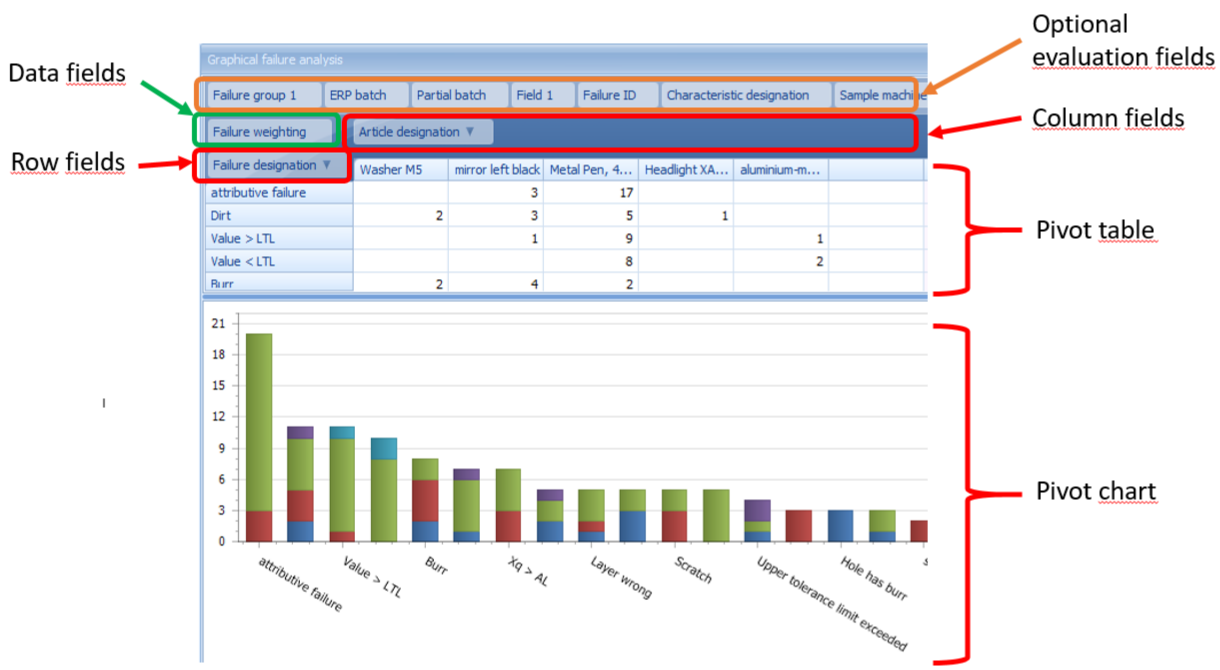
Pivot tables aggregate extensive data volumes and present a clear overview. They are the basis for simple and flexible evaluations.

Many HYDRA evaluations are based on tables and provide a parallel visualization of the data in pivot charts.

A pivot table is structured as follows:

- Row fields (row header)
- Column fields (column header)
- Data fields

Row and column fields group the data contents. The data fields provide the content of the pivot table.



There are different aggregate functions to calculate the content of the data fields. The aggregate functions are called as follows:

- Move the mouse pointer over a data field (in the screenshot field with green frame, *Failure weighting*).
- Right-click
- Select the menu item *Summary function*

The following fonctions are available to aggregate data:

- Sum (total)

- Quantity
- Minimum
- Maximum
- Average (mean value)
- Corrected sample variance
- Variance
- Corrected standard deviation
- Standard deviation
- Ratio

The different aggregate functions are identical to the general functions of pivot tables. For this reason, only the most frequent aggregate functions are described below.

To explain the aggregate functions, the data of the pivot application *Failure mode analysis* is used as an example. Data field is the field *Failure weighting*. The content of the field *Failure weighting* is the number of parts with a specific failure. The column field (column header) is the article designation and the row field (row header) is the failure designation.

Failure designation	Failure weighting
Article designation: metal pen 4-color 10000m capacity	
attr. failure	1
attr. failure	1
attr. failure	1
attr. failure	1
attr. failure	1
attr. failure	2
attr. failure	3
attr. failure	7
Article designation: Mirror left black	
attr. failure	1
attr. failure	1
attr. failure	1

Sum (total)

For each combination of row and column field, the system uses the basic data to calculate the **total** of the values for this data field.

Result of the example:

Failure weighting	Article designation		
Failure designation	Washer M5	mirror left black	Metal Pen, 4-colour…
attributive failure		3	17

Quantity

For each combination of row and column field, the system uses the basic data to calculate the **quantity (number)** of data records that include a figure for this data field.

Result of the example:

Failure weighting	Article designation		
Failure designation	Washer M5	mirror left black	Metal Pen, 4-colour…
attributive failure		3	8

Minimum

For each combination of row and column field, the system uses the basic data to identify the data record with the **smallest value** for this data field.

Result of the example:

Failure weighting	Article designation		
Failure designation	Washer M5	mirror left black	Metal Pen, 4-colour…
attributive failure		1	1

Maximum

For each combination of row and column field, the system uses the basic data to identify the data record with the **greatest value** for this data field.

Result of the example:

Failure weighting	Article designation		
Failure designation	Washer M5	mirror left black	Metal Pen, 4-colour…
attributive failure		1	7

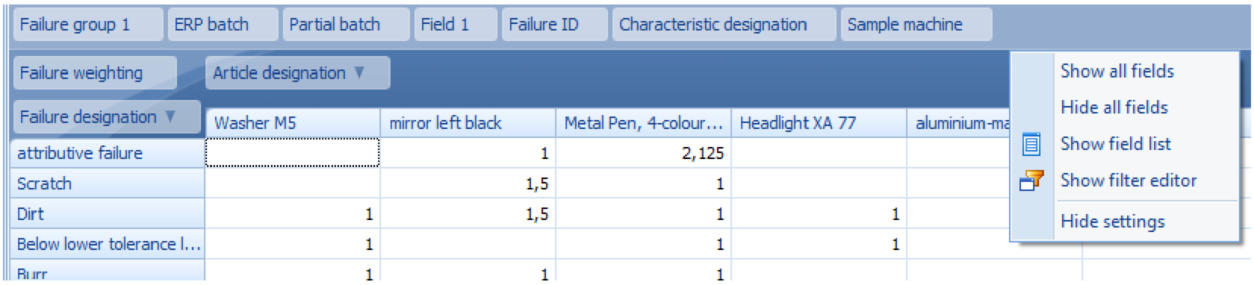
Average (mean value)

The system uses the basic data to identify the number of data records for each different combination of row and column field. The result of the aggregate function *Sum* is then divided by the number of data records identified using the basic data and is output as **average (mean value).**

Result of the example:

Field list

The area of the optional evaluation fields can be extended using fields of the field list. To activate the field list, right-click in the area of the optional evaluation fields. The following context menu opens.



Select the item *Show field list* to open a list of all available evaluation fields. Use drag and drop to drag the required fields into the area of the optional evaluation fields. You can also move a field from the field list directly into the area of the row, column or data fields. And vice versa, you can also use drag and drop to move a row or column field into the field list. Also use drag and drop to move the row and column fields into the area of the optional evaluation fields. To move a field into the field list, you must drag the field outside of the area of the pivot table.

Use the menu item *Show all fields* to move all fields of the field list into the area of the optional evaluation fields. And vice versa, use the menu item *Hide all fields* to move all fields from the area of the optional evaluation fields into the field list.

Filter editor

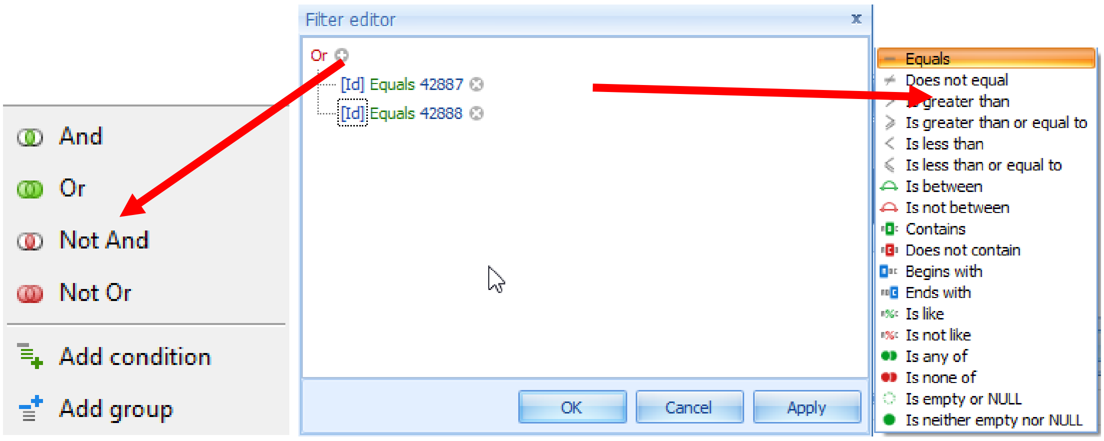
The function *Show filter editor* complements the selection criteria and helps to further narrow down the data basis. You can configure different filters and combine different parameters. Click a parameter to open the list of available parameters.

Example:

Objective = display all failures for article "42887" or "42888".

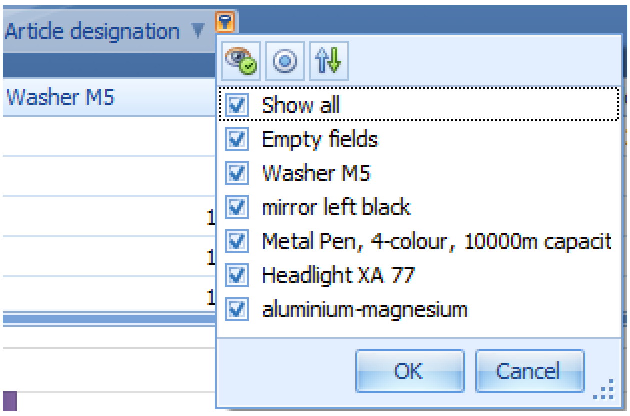
Proceed as follows:

1. Click "and" --> select "or"
2. Click the "+" next to "or"
3. Click "Finished on/at" --> select the field "article number"
4. Click "Begins with" --> select "Equals"
5. Click "Enter value" --> enter "42887"
6. Click the "+" next to "or" (on top)
7. Repeat steps 3 to 5 with number "42888"
8. Click button *Apply*.



Field filter

Use the row and column fields to filter the data of the pivot table. You can also use the fields in the area of the optional evaluation fields for filtering. If the mouse is moved over one of these fields, a small filter symbol is shown. If you click the filter symbol, a list of all data for this field opens. You can activate/deactivate specific data in this list.



Using the context menu of the data fields, you can also activate an ascending or descending sorting of the values of the relevant data field. If you click the relevant row or column field, you can also select between an ascending or descending sorting. Even after each row or column of the pivot table you can activate an ascending or descending sorting. Right-click a row or column header to activate the sorting. In the context menu that opens via right-click, you can also cancel the sorting.

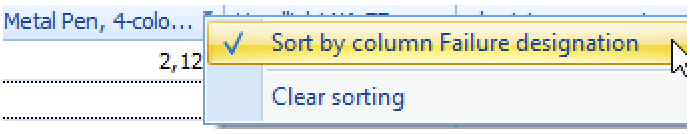
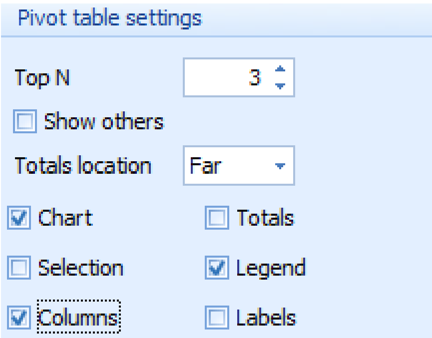


Chart settings

The menu item *Show settings* activates a setting dialog to configure the pivot chart.



Top N

The option *Top N* restricts the data displayed in the chart to the contents of the row or column fields with the "N" greatest values.

Using the context menu of the data field (*Failure weighting* in the example), you can also activate of deactivate the display of the "Top N" without opening the settings dialog for the chart.

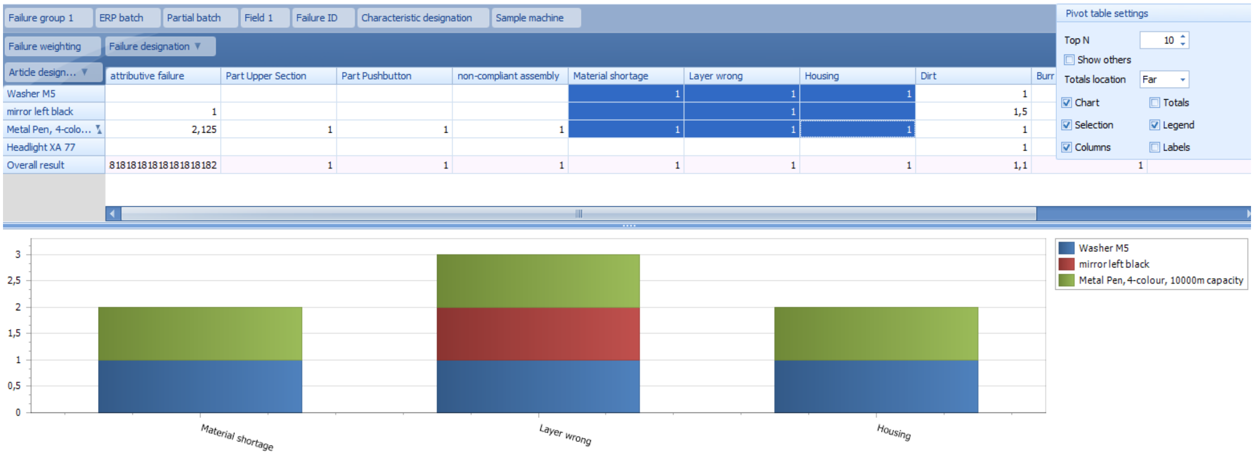
Totals

Check the *Totals* option to display the row *Overall result* in the pivot chart. If this option is enabled, the selected cells are used to calculate the overall result.

Selection

Check the *Selection* option and select a specific area to specify the contents of the tabular display. If the option *Selection* is enabled, the graphic display is based on the selected cells. If you check the *Labels* option, you can display the total number for each bar.

The screenshot below illustrates these functions.



Columns

Check/uncheck the *Columns* option to switch between the graphic presentation of the relevant number of columns or rows.

3.3.5 Print application contents

You can print all applications included in the system. The following options are available:

- *Print all*
  Use this option to print all detail applications at once.
- *Print preview*
  Use this option to print a specific detail application. Use this option, e.g. to print out a chart in a combined evaluation/report.

Both options provide the print preview. The print preview allows you to adjust printing according to your specific requirements.

Use the user/group authorizations to disable the print permission.

3.3.6 Export application contents

You can export table contents into other formats like Microsoft Excel or PDF. You can start the Excel export directly from the context menu of the table header. Once you have selected the option, the *Save as* dialog of the operating system is shown, where you can select the storage location and the file name of the document.

If you would like to export table contents into the PDF format, first select the print preview of the detail application. Click the *Export* option in the print preview to start the export into the PDF format. Once you have selected the option, the *Save as* dialog of the operating system is shown, where you can select the storage location and the file name of the document.

Use the user/group authorizations to disable the export permission.

3.3.7 Shortcuts

Enter Requests data in evaluations and overviews.

Ctrl-F4 Closes the current window (no editing dialogs).

Esc Closes an editing dialog after the security prompt/confirmation prompt.

Alt-F4 For editing dialogs: Closes the dialog after displaying a security/confirmation prompt. Otherwise: closes the application.

Ctrl-Tab Changes the current window.

Tab, ⇑-Tab cursor goes to the next or previous field .

, ↓ Scrolls through the table row by row, the selection bar is positioned in the next row each.

, ↓ In date fields, use these keys to increase or reduce the date by one day, month or year; it depends on which part of the date is selected.

← Deletes input in selection lists.

Use the user/group authorizations to disable the permission for using CTRL+C (copy).

4 System functions

4.1 Configure the MOC user interface

4.1.1 Change the MOC layout

The MOC is delivered in a pre-defined design (skin). We also deliver a selection of other skins. Consequently, you can choose a display format that meets your individual requirements.

You can configure the MOC skin in the MOC configuration that can be opened in the system menu: *Extras → Configuration*. Please note that skins change the structure of user interface elements (controls).

4.1.2 Change number, time and date format

You can change the MOC formatting options according to your requirements. To do so, open the system menu: *Extras → Configuration* and choose the required formatting options. The options offered there depend on the operating system in use and the relevant formats for numbers, times and dates apply for the selected regional settings.

4.1.3 Change the language

You can change the MOC system language. To do so, open the system menu: *Extras → Configuration* and select the required language. Please note that you can only select those languages that are licensed in the system.

4.1.4 Change the type of table export

The MOC can export table contents to Excel. In addition to the current Excel versions, the MOC also supports the formats for Excel 97-2003. Normally, the MOC automatically identifies and uses the version of the installed Office version. However, you can also set the required format manually. To set the format, open *Extras --> Configuration* in the system menu and select the required format in *Export tables as.*

If you want to perform an export to computers where no Office package is installed, you must explicitly specify the format you want to use.

If MS Office is not installed and the table export type is set to "automatic", the export function is not available.

4.1.5 Change the start menu

You can change the used MOC menu. You can find the available menu variations in the system menu in *Extras → Configuration.* Use the menu editor to create individual menu variations.

Select the *Use standard menu* option to use the default menu. This configuration is used for service purposes.

4.1.6 Change the *open mode* for applications

You can specify if applications are opened only once or if several instances of an application should be opened simultaneously when you open an application via the menu or transaction code. To do so, select the required options in: *Extras → Configuration → Application mode.*

When you call an application from another application, the target application will always be opened in a new window.

4.1.7 MDI mode

Use the MDI mode to specify if applications are managed as separate subwindows (mode *MDI window*) or as tabs (mode *tab*) in the MOC program window.

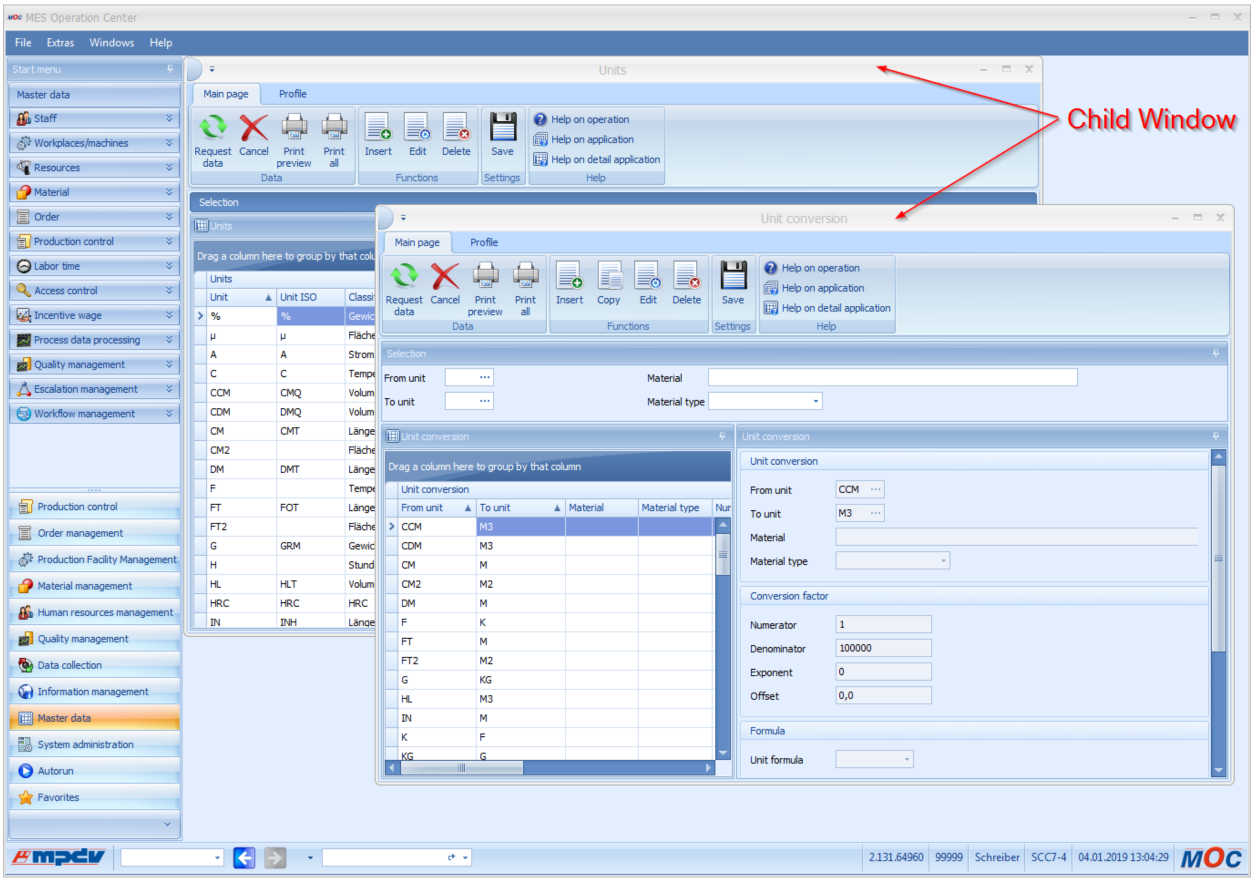
Note

Activate MDI mode option

The menu item to select the MDI mode is only available if the function authorization "mdimode.edit" is activated.

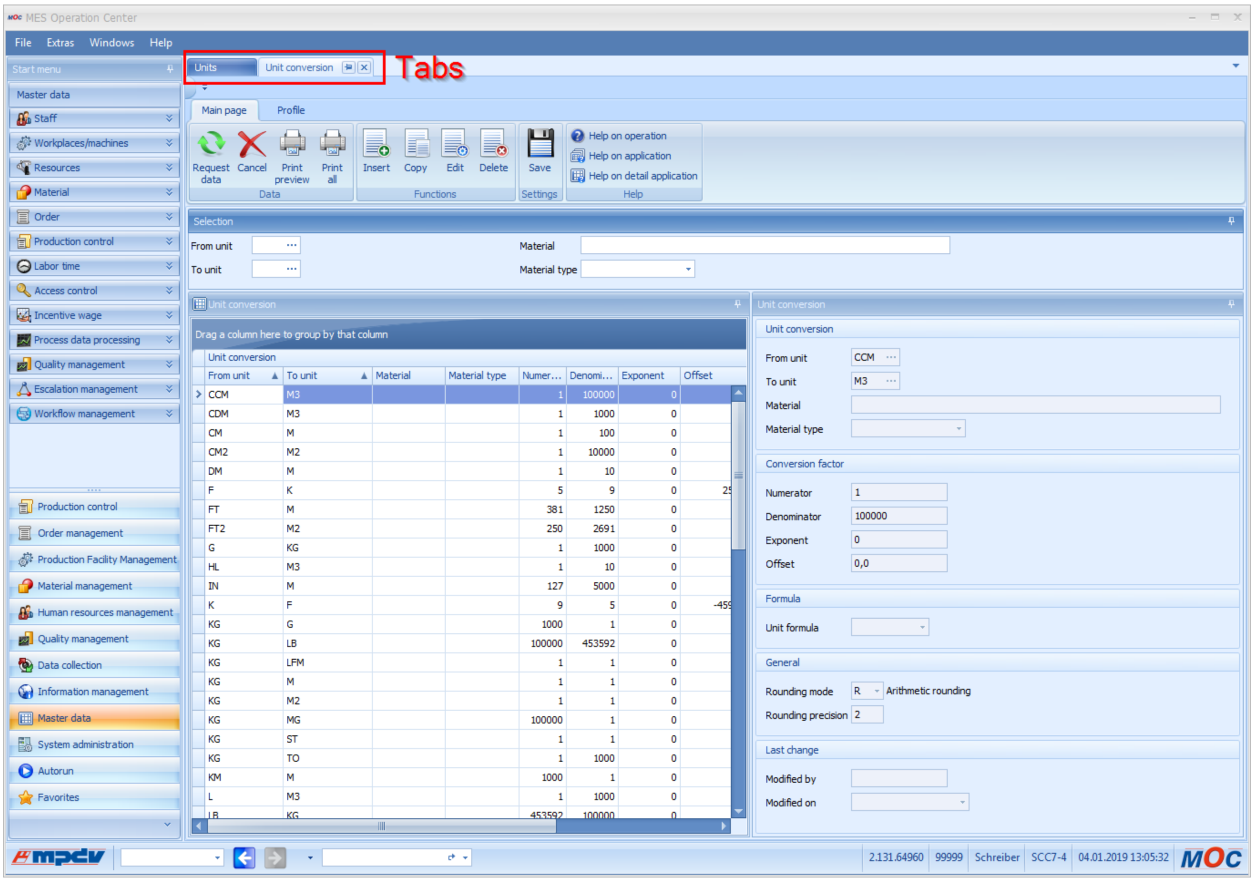
4.1.7.1 MDI window mode

If you select the *MDI window* mode, subwindows are like program windows. You can move them anywhere and change their size.



4.1.7.2 Tab mode

If you select the *Tab* mode, the system manages the subwindows as tabs. Use drag and drop and the docking options to position subwindows within the MOC program window. Additionally, you can also use applications as separate windows outside the MOC program window, e.g. on a second screen.



The selection of the *MDI mode* or the *Tab mode* is only available,if the extension mdimode_active is activated.

4.1.8 Automatic update

You can specify here if you want to search automatically for updates. If the mode *Find and import updates automatically* is enabled, the system searches for updates every time you start the program. If you select the *disabled* option, the system does not search for updates automatically. You must select this mode if Citrix is used. For information on the automatic updates, see section 4.10.2.

4.1.9 UAC mode

The UAC mode specifies if further authorizations are required to install updates. The option *automatic* is set by default. This mode checks if updates are installed in the Windows program directory. If this is the case, additional authorizations are necessary. If the *active* option is selected, the update is always executed with extended authorizations. If the *disabled* option is selected, the update is performed with the user's authorizations who started the MOC. If this user does not have sufficient authorizations, the update cannot be installed and the process is cancelled.

4.1.10 Change the type of date input

You can enter date values with or without separators in the MOC.
If you use separators to enter the date (set by default), you must use another key to switch to the next date range, e.g. enter a separator or use the <right arrow> key, after you've entered a value for day, month, year or time.

If you enter dates without separator, you switch automatically to the next date area, once the current area (day/month/year) has been completed. Note: If you use this input type and enter a separator, you also switch to the next date area.

Note: This is a global setting and applies to all date inputs on the MOC.

4.2 Menu editor

Use the menu editor to design menu variations according to your requirements. You can change existing menus, create new menus or delete your customized menus. You need the function authorization "sysmenu" to open the menu editor. Start the menu editor in the system menu: *Extras* → *Menu editor.*

Load a menu into the configurator.

If you want to change an existing menu, load it into the menu editor. To do so, select the required menu in the combo box and choose the "load" function. As a result, this menu is shown in the configurator. To open the menu, click the "+" character in front of the top menu entry.

Edit a menu

Once you have loaded a menu into the configurator, you can edit the menu by dragging individual entries from the overview list of functions to the menu (drag&drop). You can move existing entries within the menu. Please note that you can position menu entries only on the level of sub menus. In case you have not defined sub menus and you drag a menu entry to a category, the system generates a sub menu automatically. You can still change the submenu labeling at a later point in time.

Create new menu

You create a new menu by entering the name of the menu you want to create in the combo box of the menu selection. If you enter a name already used for another menu, this menu will be overwritten.

The system internally manages your individual menus with reference to your user name. This means: If you edit an existing menu, which has not been generated by the logged in user, the system internally generates a copy with reference to the user name. You then edit the user-specific copy. If you delete your copy, the system uses the original version of the menu.

Delete your customized menu

You can delete your individual menu by selecting this menu in the combo box and clicking the *delete* button. This deletion process initiated by the user deletes the locally saved menu. The other menus applying to the whole system are not affected.

4.3 Professional mode

The *Professional mode* is no longer available in MW 3.1 from service pack 12 onwards.

- Most of the functions provided by the professional mode are now integrated as "configuration" and are available without the *Professional mode*.
- Some of the functions are integrated as "customization" and are available with the relevant development license for the MES Development Suite. The category "customization" includes functions to create new data objects in the fields "dialog configuration" and "MLE configuration". If these functions are not available on the MOC, the tooltip or the online help can inform you which license is required.

In MW 3.0, the *Professional mode* is still available because the license model of the MES Development Suite was changed in the early days of MW 3.0 and not all customers with a "customization" authorization have activated the licenses in their system that are used today.

The professional mode is used for a specific system customization. But to use the functions, specific knowledge is required and the changes made imply a modification of the system. For this reason, only use these functions if specific conditions are fulfilled. You can activate the professional mode in the system menu: *Extras* → *Professional mode.*

4.4 Development Suite

The Development Suite is a development environment that can be used by authorized and trained users to create their own applications in the system. You can activate the Development Suite in the system menu in: *Extras* → *Development Suite.*

4.5 Log window

The log window shows intrinsic operations in tabular form. Only use this function on instruction for service purposes. You can activate the log window in the system menu in: *Extras* → *Log window.*

A new window opens showing the entries of the current log file. The log window dynamically reloads new entries generated by the MOC.

If you enable the *tailing* option, the tool automatically scrolls to the latest entries. This function is also enabled if you scroll down to the end of the table. If you scroll up the table, *tailing* will be disabled.

Use the option *Clear log file* to delete the contents of the current log file. Only use this function carefully. This function is useful if you want to reproduce and record a specific behavior.

Use the shortcut *CTRL + F* to open a search form you can use to browse log contents. If you use this function in combination with the filter row of the table view, you can quickly find specific log entries.

4.6 Delete user settings

The system saves the user's changes made to the application, such as the column width of tables or the alignment of applications on the desktop, in the user settings. You can delete these settings to reset a user's settings to the default factory settings. In an intermediate dialog, the system shows at first the directory from which data is to be deleted. You delete all of a user's settings in the system menu in: *Extras → Delete user settings.*

4.7 Delete user settings for the application

Use this function to delete the settings of an application. In this context, only the settings of the active application are deleted. In an intermediate dialog the system shows at first the application name and the directory from which the data is to be deleted. Delete the settings for the active application in the system menu in: *Extras → Delete user settings of the application.*

4.8 System information

The system information in *Help → System information* provides you with an overview of the system's different settings and statuses. System information is usually used for service purposes. However, some information may also be interesting to the user, such as the transaction codes. You can request the following pieces of information in the single sections of the system information function:

System

The *System* section shows the system's configurations, such as the installation location and current information, like logged in users.

Authorization

The *Authorization* tab shows the functions that may be authorized in the system as well as their authorization status.

Configuration

The *Configuration* tab shows the local configurations of the executed client instance.

Transaction codes

Use the *transaction codes* to start applications from the taskbar. The *Transaction code* tab shows which transaction code is assigned to which application.

Applications

The tab *Applications* shows the version details for each installed application.

Assembly versions

This tab shows the version of the installed system components.

4.9 Search for updates (HYDRA installation version lower than SP9)

Use the system menu function: *Help* → *Search for updates* to update your local MOC installation.

4.9.1 Overview

Normally, it is sufficient to check in the dialog whether newer updates exist on the server by clicking the *Search for updates* button. If this is the case, updates are downloaded automatically and can be installed by clicking the *Install* button. In this context, the MOC is finished automatically and an external maintenance program ("Updater") is started. Click the *Start copy process* button to start updating your installation based on the downloaded files. Click the *Start MOC* option to restart the system, once the copy process has been finished.

You can also update the MOC directly, i.e. without using an update server. To do so, you must enter the directory including the updated version in the *Update directory* text field and click the button *Import updates.*

4.9.2 Detailed description of the update process

The update process provides different options that are described in the below paragraphs.

***Search for updates* dialog**

- The *Search for updates since* option specifies which files are to be downloaded from the server. In this context, the date is compared with the change date of the files on the server. The date is updated automatically once updates have been downloaded successfully.
- Use the *All updates* option to download all files available on the server.
- The *Update server* option includes the address of the server that provides the updates. By default, the field is populated with the values specified when the MOC was installed or with values transferred from the MOC configuration file. Make sure spelling is correct for any changes you make.
- The *Search for updates* option connects to the server, searches for updates and downloads these updates, if necessary. This process might take a while, subject to the network connection and the amount of data to be downloaded. The downloaded files are then stored in the *Update directory* folder.

- Use the *Direct update import* option to start the external maintenance program automatically after downloading the files.
- The *Update directory* option includes the directory that contains the updated MOC program files. Usually, this directory receives its data from the *Search for updates function.* But you can also copy data from other sources (e.g. CD).
- The *Import updates* option terminates the MOC and starts the external maintenance program.

Maintenance program *Updater*

The external maintenance program *Updater* updates the files pertaining to the local MOC installation. The *Updater* reads the data from the specified source directory and writes this data to the defined target directory.

Normally, the maintenance program makes changes to the Windows *Programs* folder. For this reason, Windows 7 shows a UAC message when starting the maintenance program to inform the user that such modification is to take place.

You may choose from the following options:

- The *Overwrite newer files* option overwrites files in the target directory, even if they have a newer date stamp.
- The *Start* button starts the copy process. In this context, all files from the source directory are copied to the target directory if these files are more up-to-date than the existing files (or if the above option is set). Files that could not be copied are shown in the section *Uncopied files.*
- The *Start MOC* button is available, once the copy process has been completed. The *Start MOC* button closes the *Updater* and starts the MOC.

Manual start of the maintenance program "Updater"

The external maintenance program *Updater* updates the MOC. Usually, the MOC starts the *MOC Updater* but you can also start the *Updater* manually. In this case, you have to specify the following parameters.

target
Target directory for the update (MOC installation directory).
source
Source directory including the files for the update. You can also use shared networks.
startwithoutasking (optional)
Starts the update without confirmation.
appname
Name of the application that is to be updated.
appexe
Application that is to be started after the update.

Example (enter in one row):

```
C:\ProgramData\mpdv\moc\updates\updater.exe
    -target='C:\Program Files (x86)\MPDV\HYDRA 8\MOC'
    -source='C:\ProgramData\mpdv\moc\updates\files'
    -appName=MOC
    -appExe='C:\Program Files (x86)\MPDV\HYDRA 8\MOC\MOC.exe'
```

4.10 Search for updates (HYDRA installation version higher than SP9)

Use the *Update* function to update your local MOC installation. This update process starts the external maintenance program *MOC Updater.* Use the *MOC Updater* to update your MOC installation.

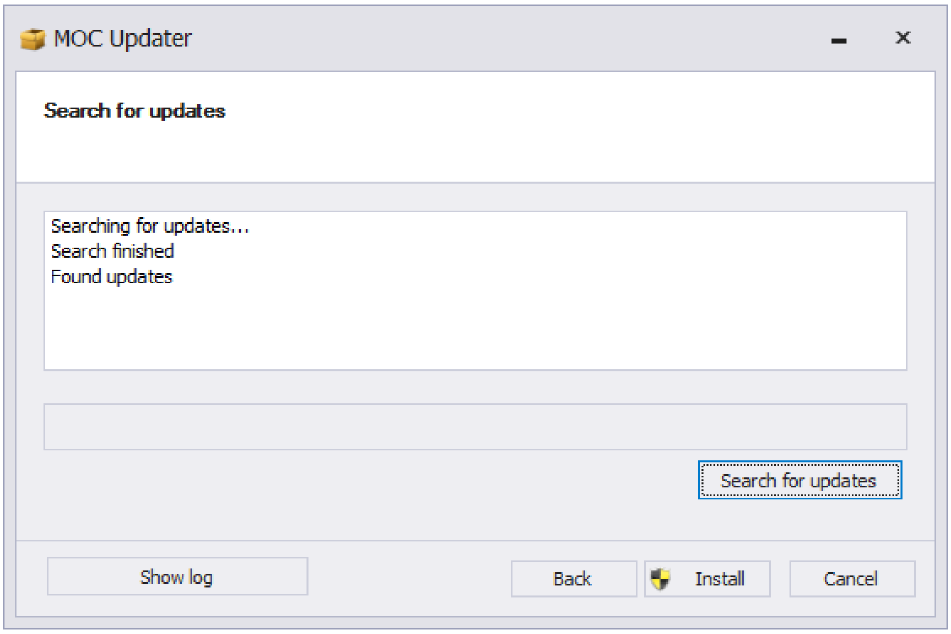
If the MOC is installed in the *Programs* directory, you need administrative permissions to write data to this directory. If you do not have these rights, Windows displays a message asking for these permissions. Optionally, you can get write permissions for the *Programs* directory. If you have write permissions but no administrative rights, you have to configure the MOC UAC settings accordingly (see 4.1.9 UAC mode).

If the update process overwrites or deletes files, the system additionally stores these files in the currently logged in user's *Local* directory. You can find these backups in the folder *MPDV\MOCUpdater\Backup\<Date><Time>.*

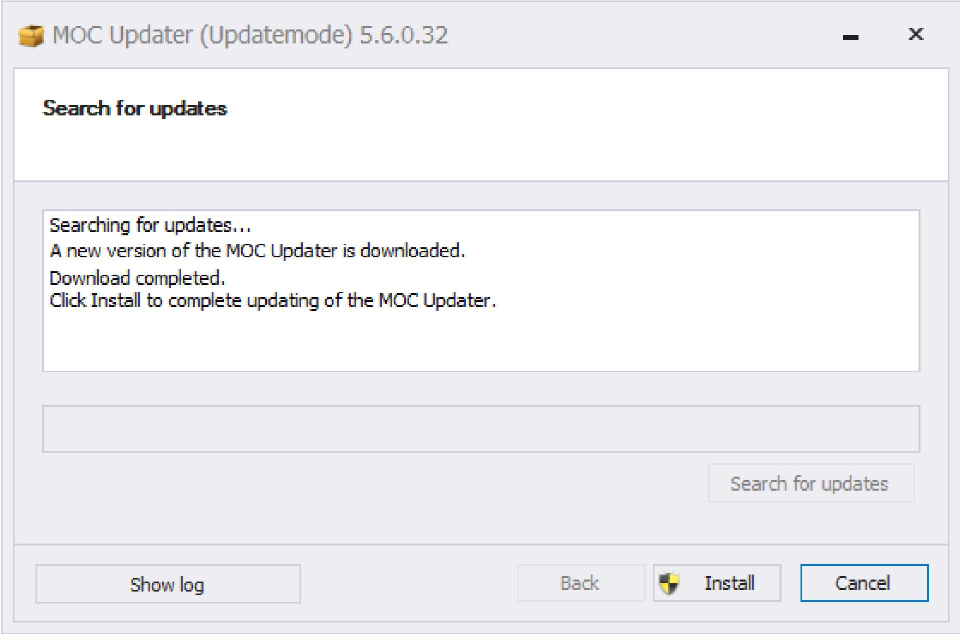
You can find further information on how to configure the *MOC Updater* in the HYDRA document dealing with the *Maintenance Manager.*

4.10.1 Manual update search

Click *Help* → *Search for updates* to start searching for updates. The search for updates is started automatically, once you have started the *MOC Updater*. You can close the *MOC Updater* if no updates can be found. Updates are downloaded if updates are found. Click *Cancel* to stop the download process.



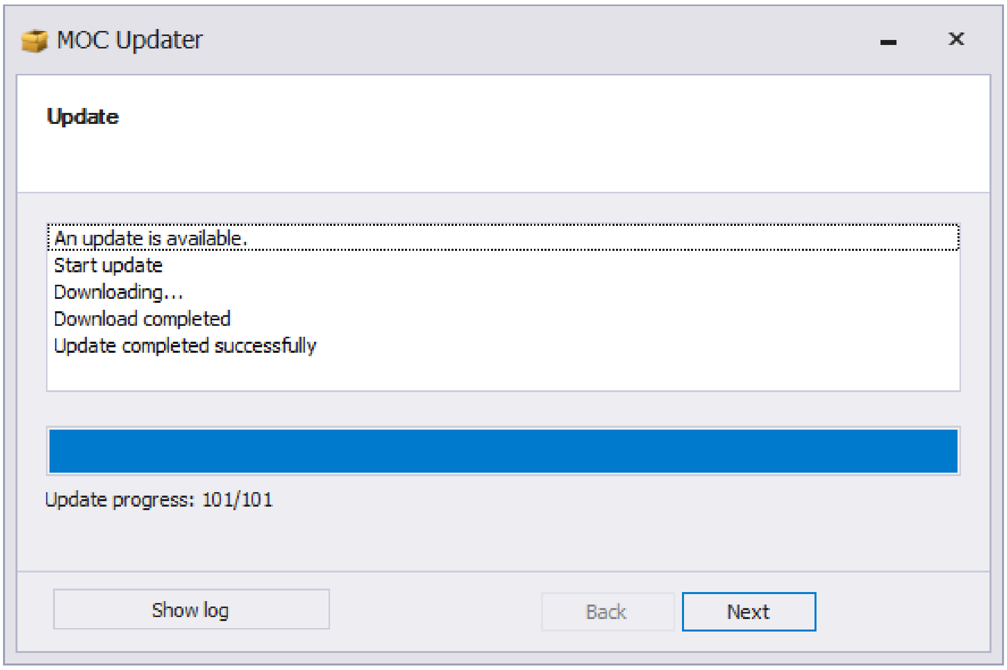
If a newer version of the *MOC Updater* is found while searching for updates, the *MOC Updater* will be updated first and then the search will be continued. Follow the instructions on the screen to update the *MOC Updater.*



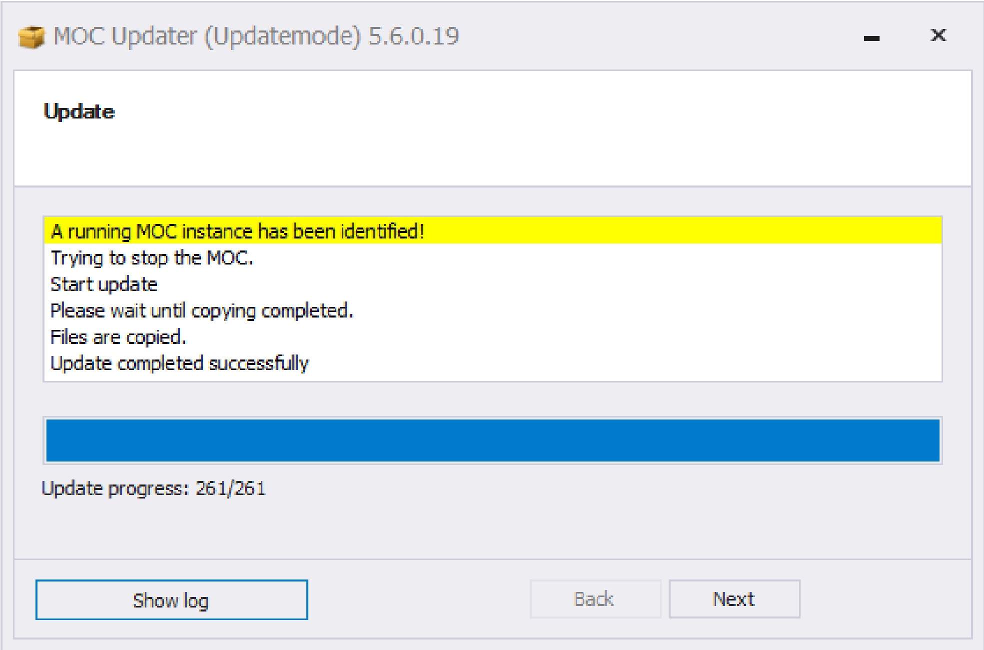
.

After downloading the update package, you can install this package. Click *Install* to continue the process. Close the MOC instance you want to update. The *MOC Updater* checks if an MOC instance is opened and, if necessary, displays a dialog window where you can close the active instance. The update process is cancelled if you do not close the instance.

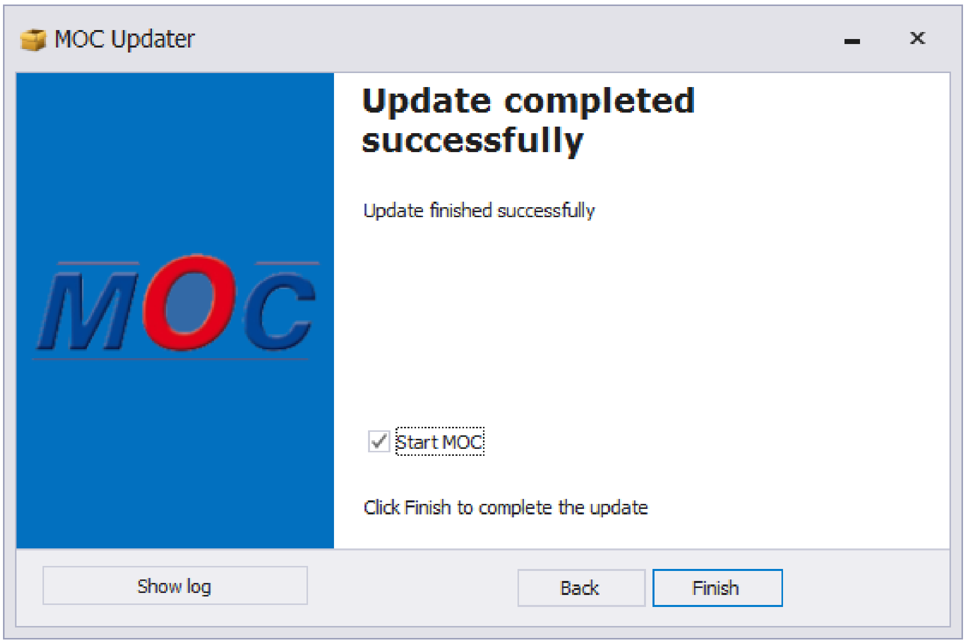
If the logged in user does not have the required authorizations to install the update, the *MOC Updater* tries to increase the user's authorization level. A message is displayed. If the authorization level cannot be increased, an error message is displayed and the process is cancelled.



Wait until the update is completed and then click *Next.*

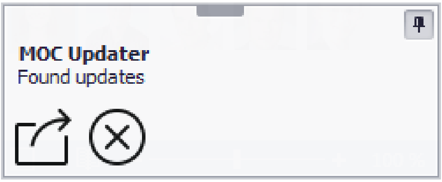


Click *Finish* to complete the update. If you do not want to start the MOC, uncheck the checkbox. Click *Open archive* to view a directory including the original files that have been changed by the update. Use these files as a backup until they are deleted after the retention period (in days) specified in *UpdateConfiguration.txt.* Click *Finish* to start the MOC.



4.10.2 Automatic update search via MOC

The automatic update search starts as soon as you log on to the MOC. The *MOC Updater* is started without active user interface. A tray icon indicates that the *MOC Updater* is active. Click the tray icon to show the user interface of the *MOC Updater.* The *MOC Updater* terminates itself if no updates are found. If updates are found, they are downloaded immediately to the computer. The following note in the taskbar informs the user about the process.



Click the message in the taskbar to show the *MOC Updater.*

After downloading the updates, the user interface of the *MOC Updater* is activated and shown to carry out the update process. See section 4.10.1 for further information on the process.

4.10.3 Manual start of the maintenance program *MOC Updater*

By default, the MOC starts the external maintenance program *MOC Updater*. The *MOC Updater* updates your MOC installation. But you can also start the program manually and configure the following parameters. The following parameters are supported:

Note that the command line to start the *MOC Updater* requires administrator rights or make sure that the required rights are available.

Parameter	**Description**
`--updateMode` **The following parameters are only effective in combination with -updateMode.**	Starts the MOC Updater in the update mode. If you do not indicate this parameter, the *MOC Updater* starts in standard mode or installation mode.
`--rootDirectory`	Target path to the MOC directory where the update process is carried out. e.g.: “c:\Program Files (x86)\MPDV\MOC\”
`--MMhost`	Address of the Maintenance Manager used for the update process. e.g.: “http://MaintenanceManager:18080“

`--searchUpdates`	Triggers the search for updates after starting the *MOC Updater.* If you add –silent, the update search is started without user interface. If updates are found, the user interface is shown. If no updates are found, the *MOC Updater* is closed automatically.
`--instantStart`	Triggers that detected and downloaded updates are installed automatically. If you add –silent, updates are installed without user interface and the *MOC Updater* is closed automatically.
`--startApplicationAfterUpdate`	Restart of the MOC after successful installation or update by the MOC Updater. As of MW 4.0pe and service pack 15, the 32-bit MOC is started by default with this parameter. Optionally, you can additionally specify `x86` or `x64` to control whether the 32- or 64-bit MOC is started. e.g.: `--startApplicationAfterUpdate x64`
`--silent`	Ensures that the user interface is not displayed during the update process. A specific tray icon indicates that the update process is running. Click the icon to activate the user interface. Additionally, the user interface is shown in cases of error and/or if user interaction is required.

`--UACmode (On/Off)`	`Off:`
	No extended authorizations are requested. The logged in user requires write permissions for the target directory.
	`On:`
	Extended authorizations are always requested.

	If the `--UACmode` is not specified, the *MOC Updater* checks if extended authorizations are needed. If the target directory is located in the *Programs* directory, extended authorizations are always required.
`--DE`	Starts the program in German (default value is English).

Example 1: Update process without user interface (enter in one row):

```
c:\Program Files (x86)\MPDV\MOC\update\MOCUpdater.exe
   --updateMode
   --searchUpdates
   --instantStart
   --silent
   --MMhost "http://hydra:18080/"
   --rootDirectory "C:\Program Files (x86)\MPDV\MOC"
```

The *MOC Updater* is started without active user interface and immediately starts searching for updates. If updates are found, they are installed immediately and the *MOC Updater* is finished.

Example 2: Update process with user interface (enter in one row):

```
c:\Program Files (x86)\MPDV\MOC\update\MOCUpdater.exe
   --updateMode
   --searchUpdates
   --MMhost "http://hydra:18080/"
   --rootDirectory "C:\Program Files (x86)\MPDV\MOC"
```

The *MOC Updater* is started with active user interface and immediately starts searching for updates. A wizard guides the user through the update process. Use this example to repair a corrupt MOC installation on a workstation computer.

4.11 Transaction codes for system functions

Use the MOC transaction codes to request various kinds of system information. The menu does not provide all of this information. The below table describes the most important functions:

Use the function *Help* → *System Information* to open the complete list of transaction codes.

sqltest	Starts the "SQL tester" application you can use to send SQL queries to the database. The function authorization "sqltest" is required. **Please note: These SQL commands are directly sent to the database where they are executed without further checking**.

sysClearCache	Clears the temporary file cache by way of which MOC accelerates access to certain files. Once the command has been executed, the cache is rebuilt automatically, which might slow down the first access to specific functions.
sysClearWSCache	Calls for server functions (web services) are buffered temporarily, if necessary, to prevent data from being called too often. If the cache is cleared, calling of this data is enforced once more before the cache time has expired.
sysclosewindows	Closes all open application windows.
sysdebugon / sysdebugoff	sysdebugon enables a special analysis mode. This mode is used when problems occur to show additional information in a status row that opens and to write this information to the log file. sysdebugoff disables (hides) the status row and reduces the number of data included in the log file.
syskeys	Shows a list of referenced but unavailable language keys as well as further potential configuration problems.
sysDebugPep	Starts the application *Workplace assignment* in a special debug mode. The debug mode remains active until the MOC is restarted.
sysDebugGrap	Starts the application *Graphic planning* in a special debug mode. The debug mode remains active until the MOC is restarted.
sysproc	Used to access running MOC background processes. *sysproc* shows running processes. *sysproc help* shows potential parameters for the command. *sysproc stop <process name>* | *all* stops the specified | all processes. *sysproc start <process name>* | *all* starts the specified | all processes. *sysproc exec <process name>* executes the specified process (once).
syssettings	Outputs a list of available system options. Use the file MOC.ApplicationSettings.config to set these values. These options control diverse internal MOC settings. Changes may only be made according to instructions or by especially trained staff.
syswslog	Opens the application to analyze the web service communication.

4.12 Allocate and deallocate licenses

You can only use the available MOC functions (applications, buttons, etc.) if the system includes sufficient licenses. A required license is only allocated, once you call the function. This can be the case, if you open applications via the start menu or if you start a function by clicking a button in an application.

In general, licenses activate a set of functions. If you start a function pertaining to a specific license, this license is allocated and all other functions assigned to this license will also be released.

The licenses allocated to a user are released once the user logs off from the system (i.e. you do not have to close the MOC).

One license can cover several functions and one function can be assigned to several licenses.

4.13 Windows

This menu item includes a list of the open applications. If you click one of the entries, the relevant application is activated.

Depending on the *MDI mode* selected (see section 4.1.7), the menu item also includes further functions to place and structure MOC sub windows automatically. Use the menu item *Reset window layout* to reset the layout and move the windows to their default place.